

Prime Modalities as Epistemic Operators

0. Purpose

We formalize the slogan “**a prime is an epistemic action**” by identifying, for each prime p , a canonical **modality package** acting on objects of a theory $\mathcal{C}$. The package decomposes any object into p -supported (fragile/obstructive) content and p -robust (inverted-away) content, with p -adic resolution supplied by completion.

The goal is to present a definition that is:

- **Mathematically anchored:** it specializes to standard constructions in $\mathbf{Ab}$, $R\text{-Mod}$, derived categories, and stable homotopy.
- **Operational:** it induces measurable information-gain functionals on ensembles.
- **Portable:** it can be represented inside noncommutative/quantum settings without claiming arithmetic itself is noncommutative.

1. Prime Modality Package (axiomatic)

1.1 Ambient setting

Let $\mathcal{C}$ be a **presentable stable ∞ -category** (or a triangulated category with compatible exactness hypotheses) admitting the relevant localizations. Fix a prime p .

1.2 Definition: Prime Modality Package

A **Prime Modality Package** at p is a triple of endofunctors

$$(\Gamma_p, L_p, \Lambda_p) : \mathcal{C} \rightarrow \mathcal{C}$$

with natural transformations making the following data and properties hold.

(A) Complementary supported/local parts. There exists a natural exact triangle (cofiber sequence)

$$\Gamma_p X \rightarrowtail X \rightarrowtail L_p X \rightarrow \Sigma \Gamma_p X$$

for every $X \in \mathcal{C}$.

Interpretation:

- Γ_p extracts the p -**supported** / p -**fragile** content.
- L_p is the p -**localization** (inverting everything not p).

(B) Idempotence (modal behavior).

$$L_p \circ L_p \simeq L_p, \Gamma_p \circ \Gamma_p \simeq \Gamma_p.$$

Typically, L_p is a **localization** (idempotent monad) and Γ_p is a **colocalization** (idempotent comonad).

(C) Completion (resolution). Λ_p is a p -adic resolution functor (completion) satisfying an appropriate universal property (e.g., right adjoint to inclusion of p -complete objects) and interacting with L_p by a canonical comparison map

$$\Lambda_p X \rightarrowtail \Lambda_p L_p X \quad \text{or} \quad L_p \Lambda_p X \rightarrowtail L_p X$$

(depending on the model).

(D) Fracture / gluing principle. There is a canonical **fracture square** (pullback square) expressing X as compatible gluing of its p -local and p -complete views:

$$X \simeq L_p X \times_{\{L_p \Lambda_p X\}} \Lambda_p X.$$

This is the formal content of “a prime provides epistemic views that reconstruct the whole.”

Epistemic reading. Applying L_p or Γ_p is an *information-restricting modality* with controlled loss; the fracture square states that the restricted views determine the original object up to equivalence.

2. Concrete models

For an abelian group A :

• **Lcalization:**

$$L_p(A) := A \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}.$$

• **p -power torsion (support):**

$$\Gamma_p(A) := \{a \in A : \exists n \geq 1, p^n a = 0\}.$$

• **Completion:**

$$\Lambda_p(A) := \varprojlim_n A/p^n A.$$

Finite primary decomposition. If A is finite,

$$A \cong \bigoplus_p \Gamma_p(A).$$

Thus primes literally index orthogonal “epistemic components.”

In $D(R)$ for a commutative ring R , one has standard models:

- L_p : derived localization $X \otimes_R^L R_{(p)}$ (or localization away from p).
- Γ_p : local cohomology / derived p -power torsion functor.
- Λ_p : derived p -adic completion (derived inverse limit).

The exact triangle and fracture square become standard “local-to-global” statements in the derived setting.

2.3 Stable homotopy / spectra

In spectra, p -localization and p -completion are Bousfield localizations; Γ_p corresponds to local cohomology-type support. The package becomes a chromatic-style epistemic decomposition.

3. Epistemic information gain (operational layer)

3.1 Prime-indexed observations

Let X be a random variable valued in integers (or arithmetic objects). Define prime-indexed observations, e.g.

$$O_{\{p,n\}}(X) := X \bmod p^n, \quad O_p(X) := v_p(X).$$

3.2 Epistemic gain spectra

Define information gain:

$$G_p(X) := \mathbb{I}(\text{bigl}(X; O_p(X)\bigr), \quad G_{\{p,n\}}(X) := \mathbb{I}(\text{bigl}(X; O_{\{p,n\}}(X)\bigr).$$

Interpretation:

- Large G_p indicates that the prime p is an *epistemically powerful* question for that ensemble.
- The family $\{G_p\}_p$ is a **prime information spectrum** fingerprinting the generative process.

This replaces any vague “entropy reduction” claim with a standard conditioning/mutual-information statement.

4. Quantum representation (prime(-power) phase space)

4.1 Prime-power dimensions

For qudits of dimension $d = p^k$, the finite field $\mathbb{F}_{p^k}$ gives a canonical discrete phase space $\mathbb{F}_{p^k}^2$. Weyl-Heisenberg operators yield clean generalized Pauli structure.

4.2 Prime-indexed channels as epistemic sieves

Define a prime(-power) epistemic channel Φ_{p^k} by averaging over a structured measurement family (e.g., stabilizer eigenbases / Clifford or MUB-like families when available). Successive application of Φ_{p^k} implements “sieving” as repeated controlled conditioning.

Caution. Claims should be *structural* (canonical constructions; easier design and analysis) rather than physical (automatic decoherence reduction).

5. Noncommutativity (representation-dependent, quantified)

Arithmetic localizations typically commute up to canonical equivalence. Noncommutativity can arise only after choosing a representation into a noncommutative algebra (e.g., observable algebra, channel composition).

Define a **commutator defect** for represented prime actions $\widehat{L}_p, \widehat{L}_q$:

$$\Delta_{\{p,q\}}(X) := \|\widehat{L}_p\widehat{L}_q(X) - \widehat{L}_q\widehat{L}_p(X)\|,$$

for a chosen norm/metric in the representation. This quantifies contextual incompatibility without attributing noncommutativity to primes themselves.

6. Multiplicity and “place sensitivity”

Multiplicity invariants in commutative algebra and geometry are defined at ideals (especially maximal ideals), i.e., at **places**. Prime modalities are exactly the operators that move an object to a place (localization/completion/support), after which multiplicity becomes visible and computable.

7. Fast validation program (minimal)

Prove:

1. $L_p(A) = A \otimes \mathbb{Z}_{(p)}$ is left adjoint to inclusion of p -local groups.
2. Γ_p is functorial, left-exact, idempotent on suitable subcategories.
3. For finite A , $A \cong \bigoplus_p \Gamma_p(A)$.

7.2 Prime information spectrum on toy ensembles

Compute G_p and $G_{p,n}$ for datasets/ensembles:

- uniform integers $1 \dots N$,
- polynomial outputs $f(n)$,

- multiplicative processes. Show spectra recover planted prime structure.

7.3 Prime-power quantum simulation

Compare $d = p^k$ vs composite d for:

- ease/canonicity of Weyl–Heisenberg and Clifford constructions,
- structured tomography sample complexity under matched assumptions.

8. North-star object: Prime Modality Stack

Define the **Prime Modality Stack** as a map assigning to each prime p (or prime power p^k) the tuple

$$\mathfrak{P}(p) := (\Gamma_p, L_p, \Lambda_p, G_p, \Phi_p)$$

where $(\Gamma_p, L_p, \Lambda_p)$ are the algebraic modalities, G_p is the operational information spectrum on a chosen ensemble, and Φ_p is a represented/quantum channel realizing “prime questioning” in a noncommutative setting.

This stack is the unified formalization of **primes as epistemic operators**.

9. References (selected)

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